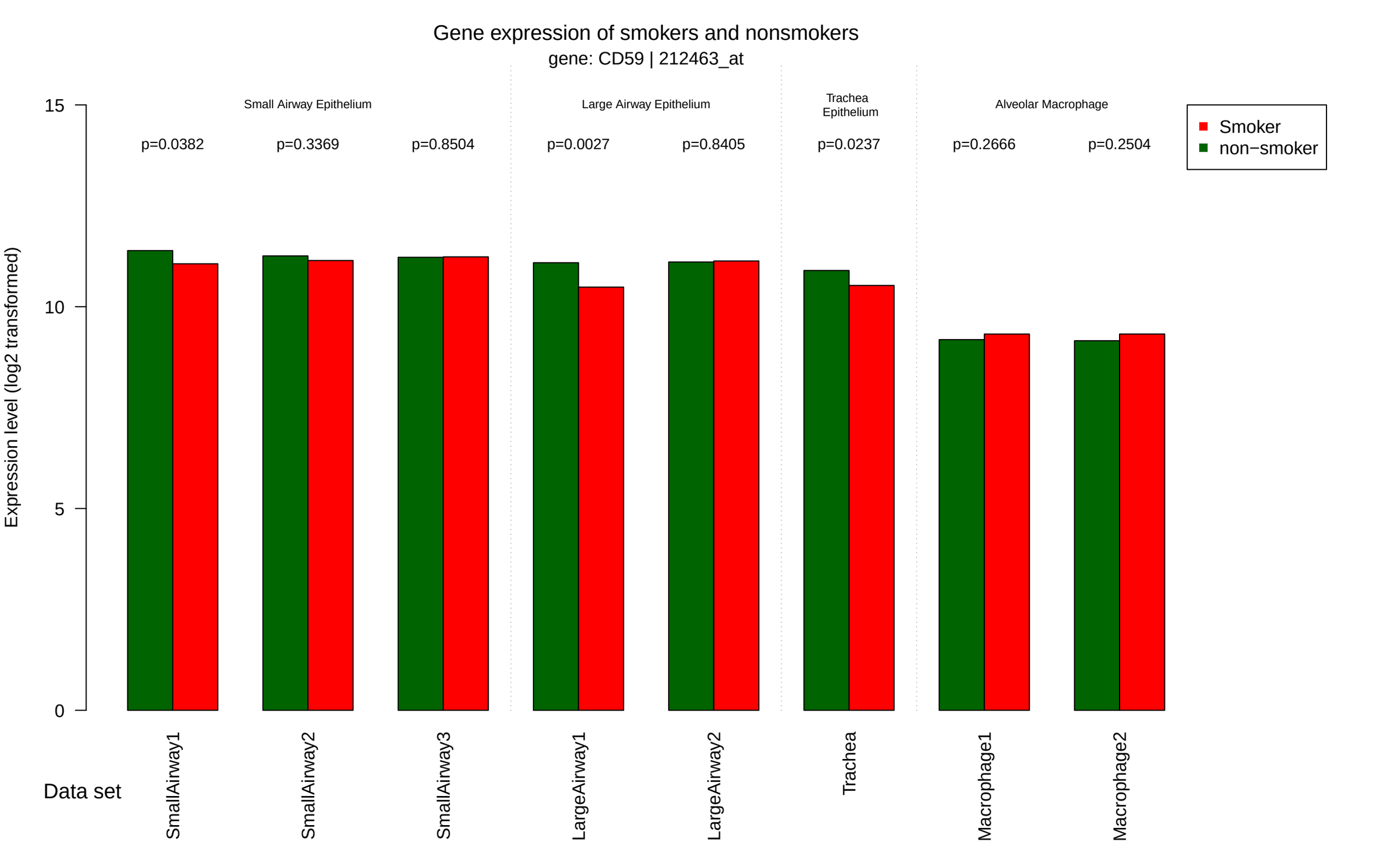
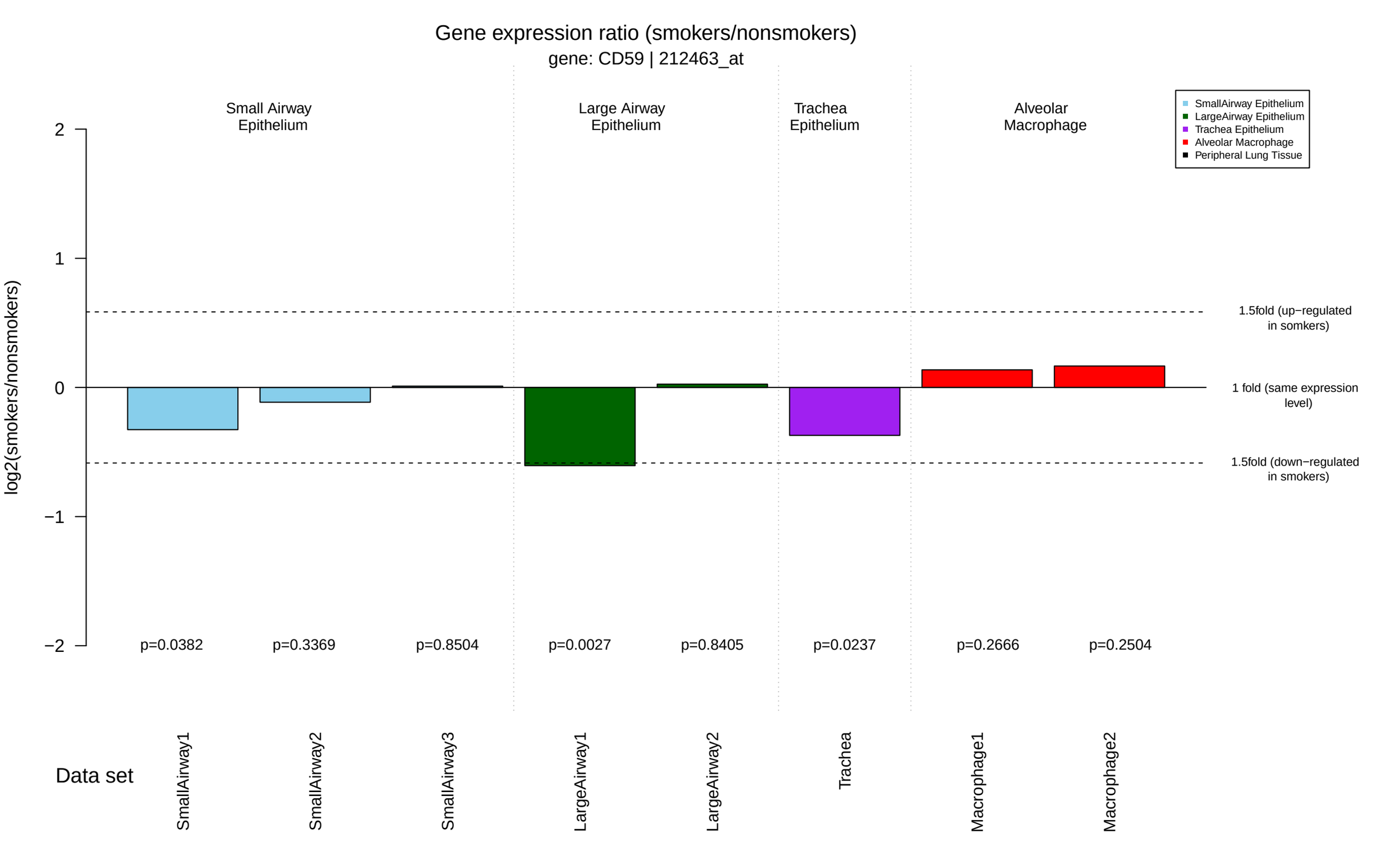
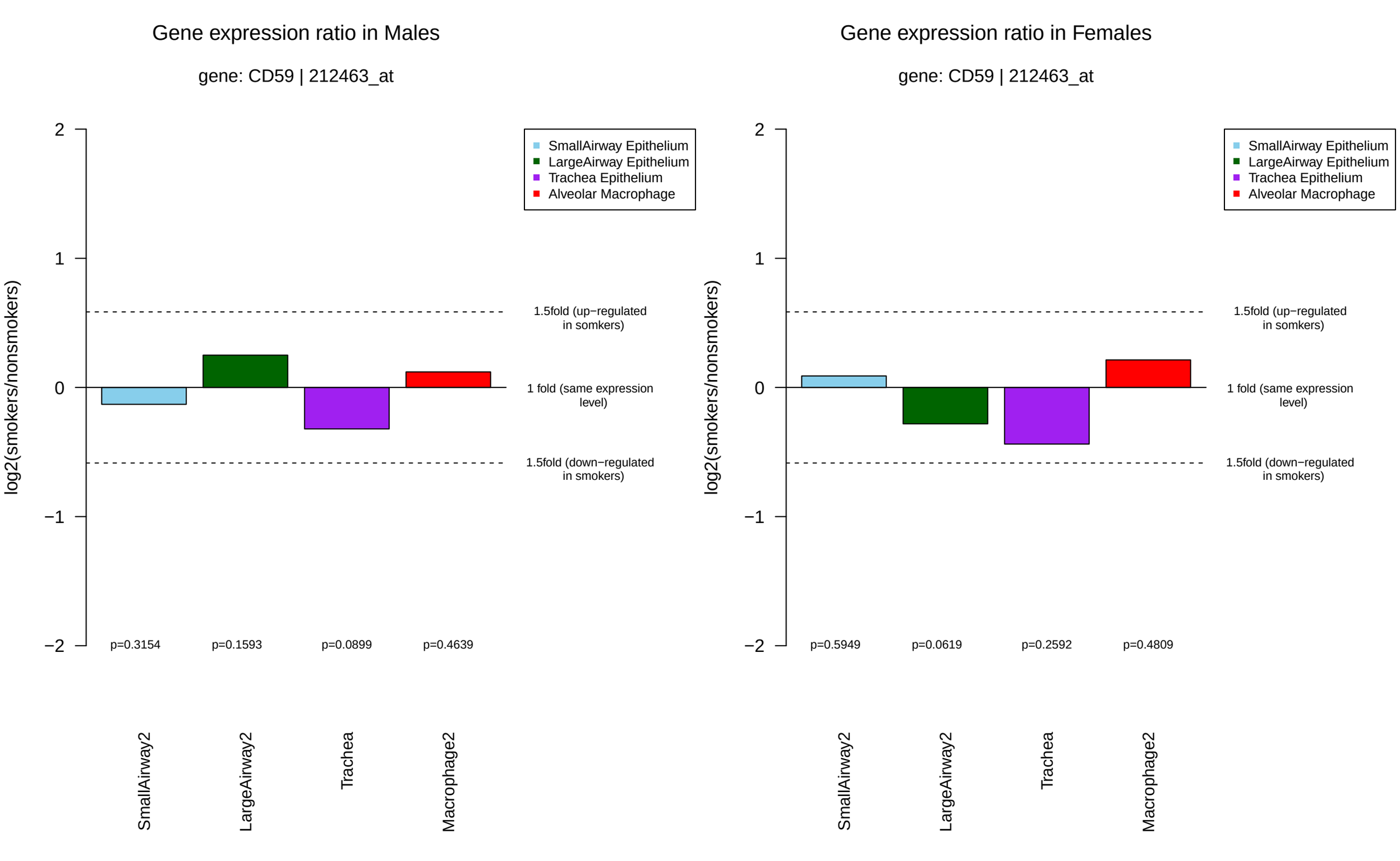
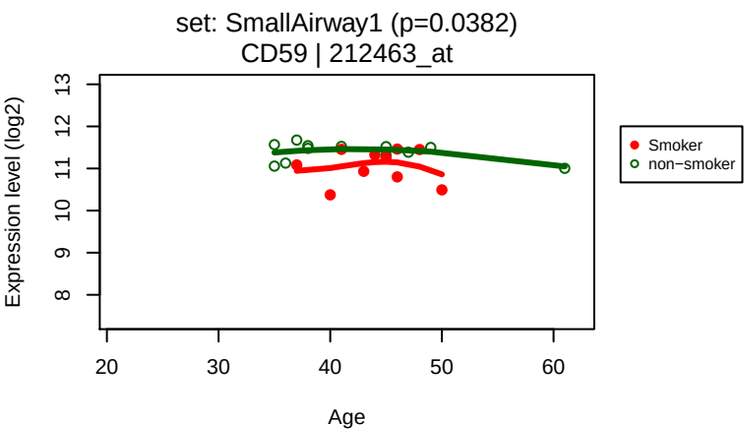
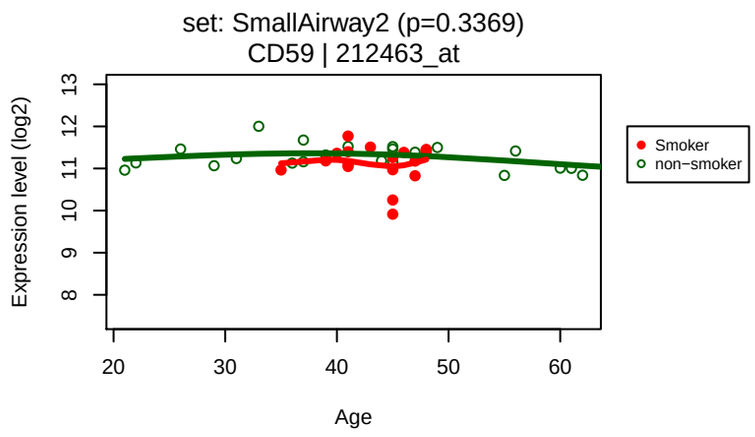
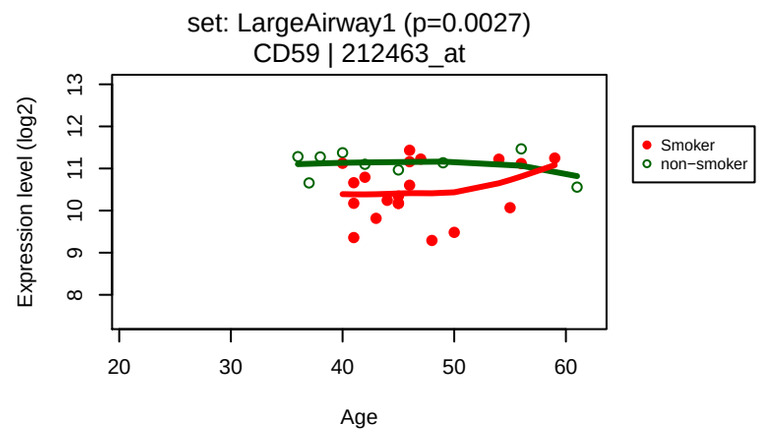

Gene expression of smokers and nonsmokers
gene: CD59 | 212463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0382
p=0.3369
p=0.8504
p=0.0027
p=0.8405
p=0.0237
p=0.2666
p=0.2504
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

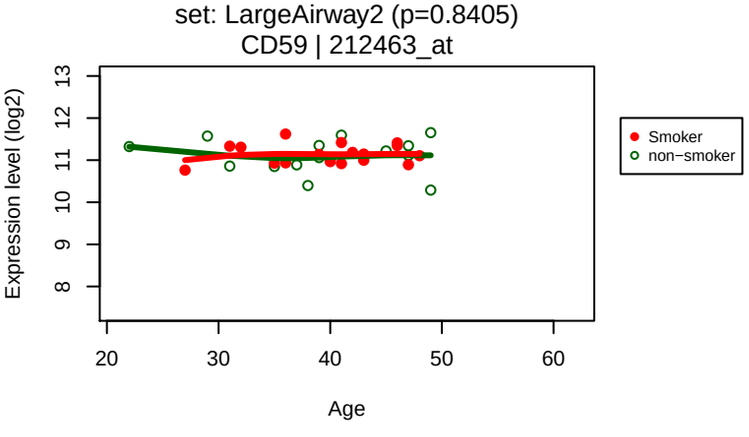

Gene expression ratio (smokers/nonsmokers)
gene: CD59 | 212463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0382
p=0.3369
p=0.8504
p=0.0027
p=0.8405
p=0.0237
p=0.2666
p=0.2504
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

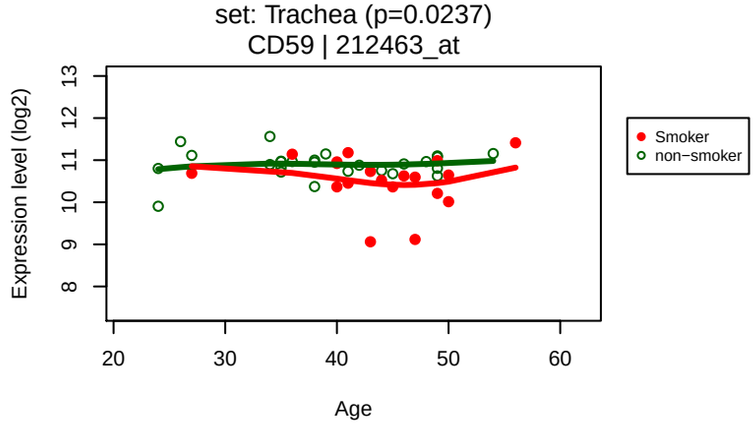

Gene expression ratio in Males
gene: CD59 | 212463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3154
p=0.1593
p=0.0899
p=0.4639
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CD59 | 212463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5949
p=0.0619
p=0.2592
p=0.4809
SmallAirway2
LargeAirway2
Trachea
Macrophage2

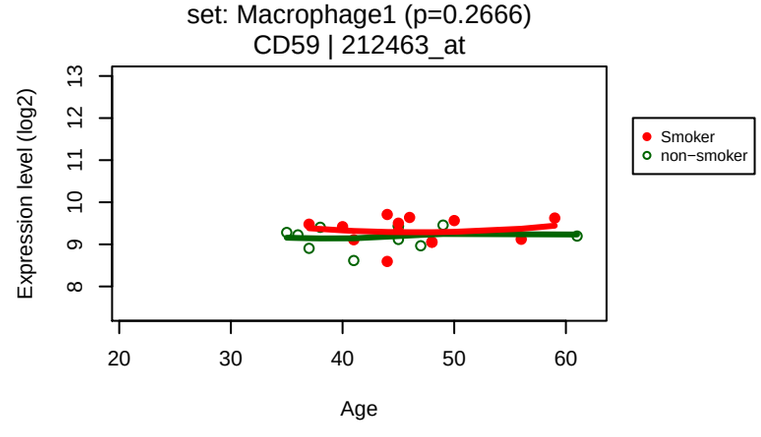

set: SmallAirway1 (p=0.0382)
CD59 | 212463_at
Expression level (log2)
Age
Smoker
non−smoker

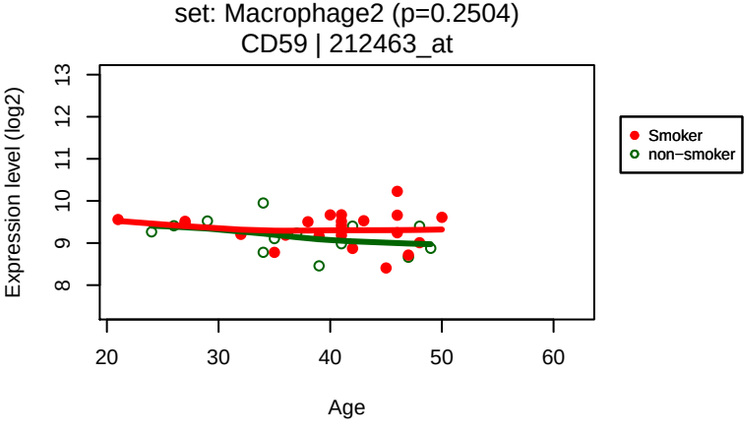

set: SmallAirway2 (p=0.3369)
CD59 | 212463_at
Expression level (log2)
Age
Smoker
non−smoker

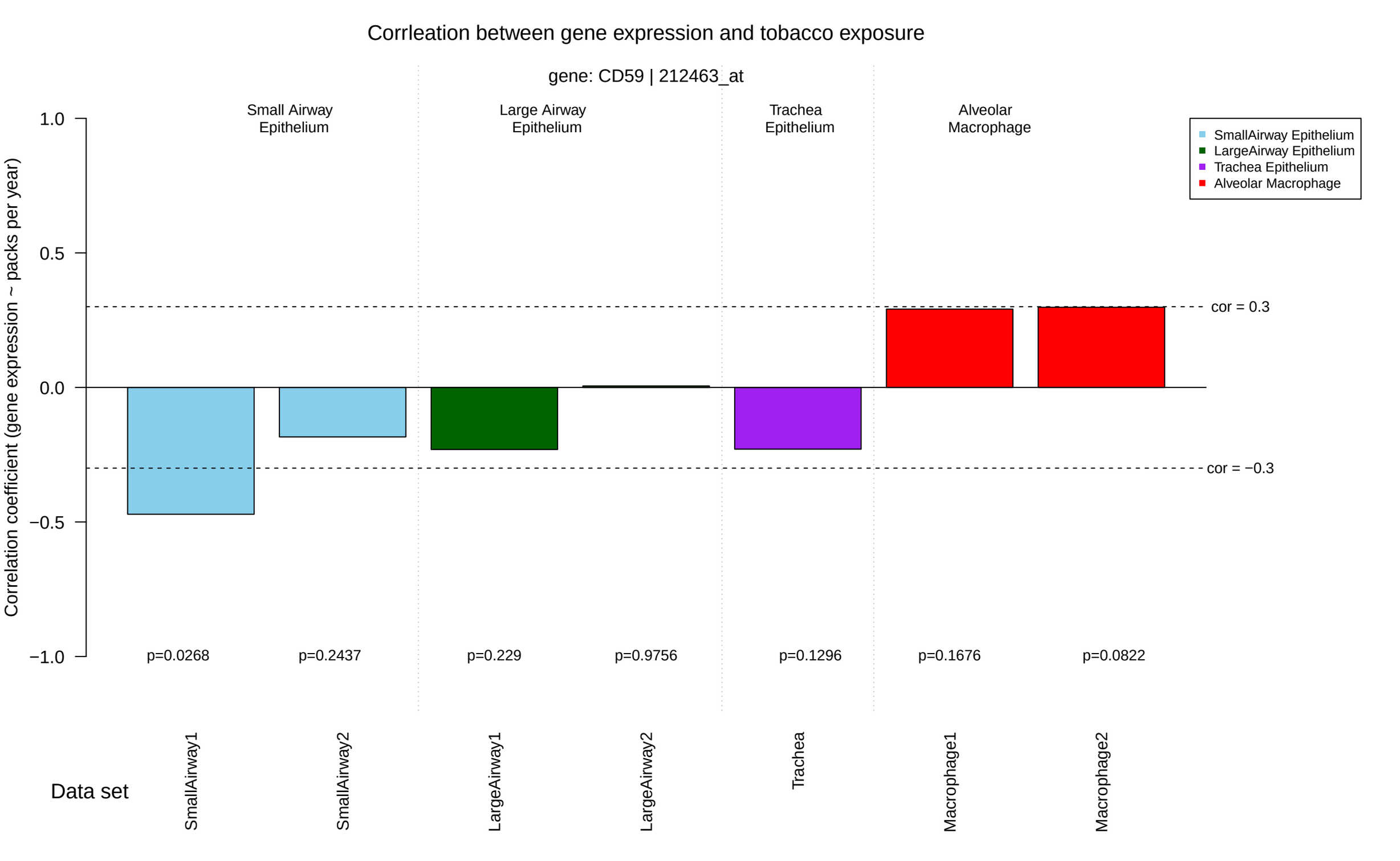

set: LargeAirway1 (p=0.0027)
CD59 | 212463_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8405)
CD59 | 212463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0237)
CD59 | 212463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2666)
CD59 | 212463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2504)
CD59 | 212463_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CD59 | 212463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0268
p=0.2437
p=0.229
p=0.9756
p=0.1296
p=0.1676
p=0.0822
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2